

MONG CAI, Viet Nam

WMO: 488380

Lat: 21.517N

Lon: 107.967E

Elev: 23

StdP: 14.68

Time Zone: 7.00 (E07)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	47.1	49.3	31.6	25.9	54.6	35.9	30.9	55.0	14.4	56.8	13.1	58.2	3.2	50	0.277

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	91.5	81.4	90.4	81.3	89.4	81.2	84.2	88.6	83.4	87.9	82.8	87.2	5.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.9	171.8	87.2	82.2	168.0	86.7	81.6	164.4	86.1	48.5	88.9	47.7	88.1	46.9	87.5	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.7	11.5	9.6	43.6	94.1	2.2	1.1	42.0	94.9	40.7	95.5	39.5	96.2	37.9	97.0
			39.4	85.1	2.4	1.0	37.6	85.8	36.2	86.4	34.9	87.0	33.1	87.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	74.0	59.7	61.8	67.3	74.7	80.3	83.4	83.6	83.3	82.0	77.7	70.8	62.9	(6)
	DBStd	9.78	5.65	6.58	5.81	4.77	3.31	2.49	2.37	2.37	2.48	3.69	5.14	5.38	(7)
	HDD50	5	3	1	0	0	0	0	0	0	0	0	0	1	(8)
	HDD65	473	179	128	45	3	0	0	0	0	0	0	13	105	(9)
	CDD50	8771	304	332	536	740	940	1003	1041	1032	961	858	624	400	(10)
	CDD65	3765	15	39	117	293	475	553	576	567	511	393	187	39	(11)
	CDH74	37825	6	51	297	1827	4852	6760	7109	6806	5717	3381	956	63	(12)
	CDH80	12977	0	0	9	276	1456	2684	2872	2629	2038	908	104	1	(13)
(14) Wind	WSAvg	3.6	3.4	3.3	3.2	3.4	3.6	3.8	4.0	3.2	3.6	3.8	3.7	3.8	(14)
(15) Precipitation	PrecAvg	103.6	1.7	1.6	2.4	4.0	9.7	18.2	25.7	18.0	10.9	3.7	3.0	1.3	(15)
	PrecMax	152.2	10.2	10.0	6.3	11.3	23.7	32.6	58.6	49.0	18.9	19.3	10.6	4.4	(16)
	PrecMin	72.2	0.0	0.0	0.3	0.3	1.9	6.6	4.8	3.7	1.3	0.3	0.0	0.1	(17)
	PrecStd	19.5	2.1	1.8	1.6	3.1	5.3	6.7	12.3	9.4	4.6	4.0	2.9	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.6	78.6	81.2	86.5	90.2	92.5	92.6	93.2	91.8	90.0	84.5	78.1	(19)
		MCWB	67.1	72.6	75.2	78.3	81.2	82.3	82.4	81.1	79.5	77.2	73.6	70.2	(20)
	2%	DB	71.7	75.5	79.1	84.7	88.3	90.8	90.8	91.2	90.4	87.6	82.6	75.7	(21)
		MCWB	65.5	70.8	73.9	77.8	80.5	81.9	82.4	81.7	79.4	75.9	73.5	69.3	(22)
	5%	DB	70.0	73.1	77.2	83.0	87.1	89.5	89.8	89.9	89.1	86.2	81.0	73.7	(23)
		MCWB	65.4	69.2	73.0	77.1	80.0	82.0	82.2	81.5	79.3	75.5	72.9	67.3	(24)
	10%	DB	68.2	71.0	75.3	81.5	85.9	88.3	88.7	88.7	87.9	84.8	79.3	71.6	(25)
		MCWB	64.3	67.8	71.8	76.4	79.3	81.6	81.9	81.2	79.0	74.9	71.7	65.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.6	73.6	76.5	81.3	84.2	84.6	85.4	84.6	83.2	80.6	78.1	74.5	(27)
		MCDB	71.6	77.8	79.8	83.8	88.1	89.4	89.6	89.3	88.2	86.1	81.6	76.7	(28)
	2%	WB	68.3	71.7	74.9	79.5	82.6	83.7	84.3	83.5	82.1	79.2	76.8	72.0	(29)
		MCDB	70.3	74.8	77.8	82.6	86.3	88.3	88.5	88.4	87.3	84.4	80.3	74.0	(30)
	5%	WB	66.8	70.2	73.8	78.2	81.6	83.1	83.4	82.7	81.4	78.3	75.6	69.6	(31)
		MCDB	68.9	72.3	76.6	81.6	85.3	87.5	87.8	87.7	86.6	83.3	79.1	72.1	(32)
	10%	WB	65.3	68.6	72.4	77.2	80.8	82.5	82.7	82.0	80.6	77.4	74.2	67.2	(33)
		MCDB	67.6	70.6	74.9	80.7	84.5	86.7	87.2	86.9	85.8	82.1	77.8	70.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	8.1	7.3	7.7	8.4	8.1	8.1	9.0	10.3	11.6	11.7	11.4	(35)
		MCDBR	11.9	10.6	9.3	9.5	9.6	9.8	9.6	11.2	12.2	13.7	13.8	13.7	(36)
	5% WB	MCWBR	7.0	6.7	5.8	5.0	4.3	3.9	4.0	4.5	4.6	5.6	6.8	7.6	(37)
		MCWBR	8.6	8.7	8.3	8.1	8.3	8.6	8.7	10.1	10.4	10.9	9.6	9.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.608	0.644	0.807	0.759	0.561	0.533	0.508	0.570	0.615	0.712	0.598	0.622	(40)
		taud	1.737	1.698	1.448	1.570	2.049	2.145	2.208	2.007	1.860	1.622	1.812	1.729	(41)
	Ebn at Noon		209	213	186	199	242	246	252	238	225	194	211	199	(42)
		Edh at Noon	68	74	99	89	55	49	46	56	65	79	62	66	(43)
(44) All-Sky Solar Radiation	RadAvg		718	812	876	1155	1413	1418	1452	1440	1421	1298	1068	837	(44)
	RadStd		179	145	130	135	134	180	171	86	91	92	152	147	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	+0.83	+0.34	+0.31	N/A	N/A	N/A	+112

Nomenclature: See separate page